

Active Directory Domain Services (AD DS) Infrastructure Lab Report

Roll Scope: Windows Server Systems Administration & Active Directory Engineering

Environment: Isolated Virtual/Physical Lab Environment

Executive Summary

This project demonstrates the design, deployment, and configuration of an **Active Directory Domain Services (AD DS)** environment running on **Windows Server**. The objective was to build a functional, centralized identity management system (*lab.local*) from scratch, resolve low-level networking and prerequisite challenges, and establish proper administrative tools and initial user directory structures.

Key Skills & Technologies Demonstrated

- **Operating System:** Windows Server
- **Directory Services:** Active Directory Domain Services (AD DS), Domain Controller Promotion (*dcpromo* / Server Manager Wizard)
- **Networking & Protocols:** DNS architecture, Loopback Adapter Configuration, TCP/IPv4 Configuration (*127.0.0.1* loopback), NetBIOS
- **Administrative Tools:** PowerShell CLI, Remote Server Administration Tools (RSAT), Active Directory Users and Computers (*dsa.msc*), MMC Snap-ins
- **Troubleshooting:** Resolving network interface binding errors, resolving DSRM/administrative credential mismatches, clearing installation prerequisite failures

Technical Implementation Steps

Phase 1: Virtual Network Interface & IP Configuration

- **Problem:** Physical network interfaces were disconnected (0 packets received), preventing AD DS prerequisites from passing due to missing active network bindings.
- **Solution:** Provisioned a **Microsoft KM-TEST Loopback** via Device Manager to simulate a persistent network interface.
- **IP Configuration:**
 - **IP address:** *192.168.1.10*
 - **Subnet Mask:** *255.255.255.0*
 - **Preferred DNS Server:** *127.0.0.1* (Self-referential loopback for local DNS resolution)

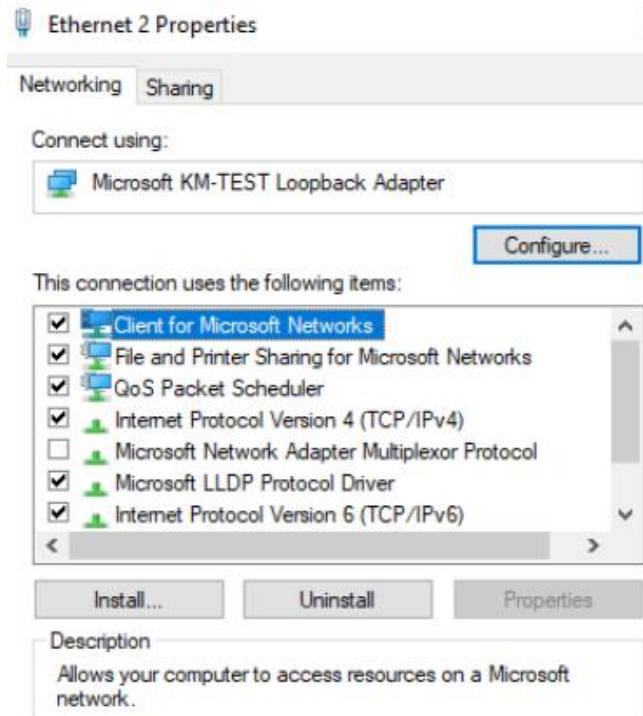


Figure 1: TCP/IPv4

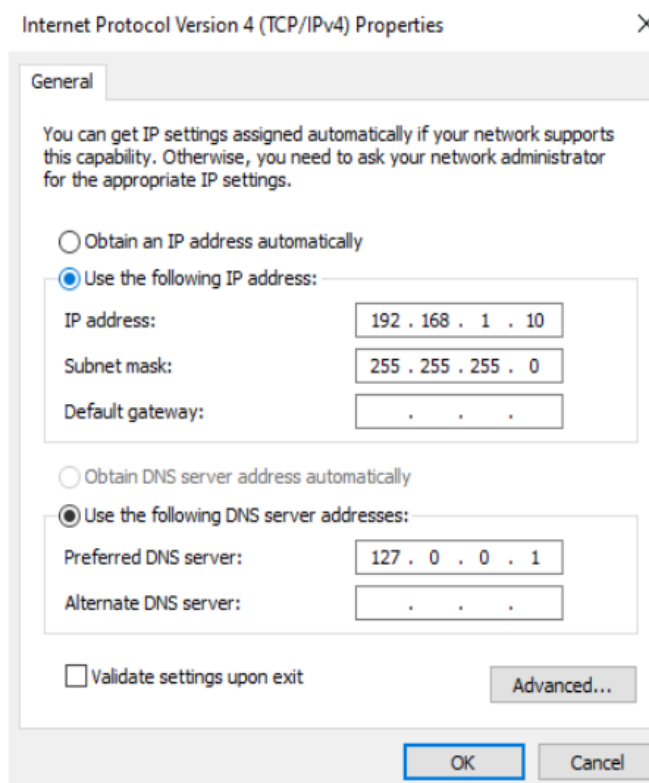


Figure 2: IP address manually

Phase 2: Active Directory Domain Services Deployment

- **Domain Identity:** Provisioned root domain *lab.local* with NetBIOS name *LAB*.
- **Server Role:** Promoted server to Primary Domain Controller (*DC-01*).
- **DNS Delegation Configuration:** Diagnosed and documented the standard “DNS Delegation” warning expected in isolated/non-integrated lab environments.
- **Credential Management:** Set up secure Directory Services Restore Mode (DSRM) credentials complying with Windows domain complexity policies.

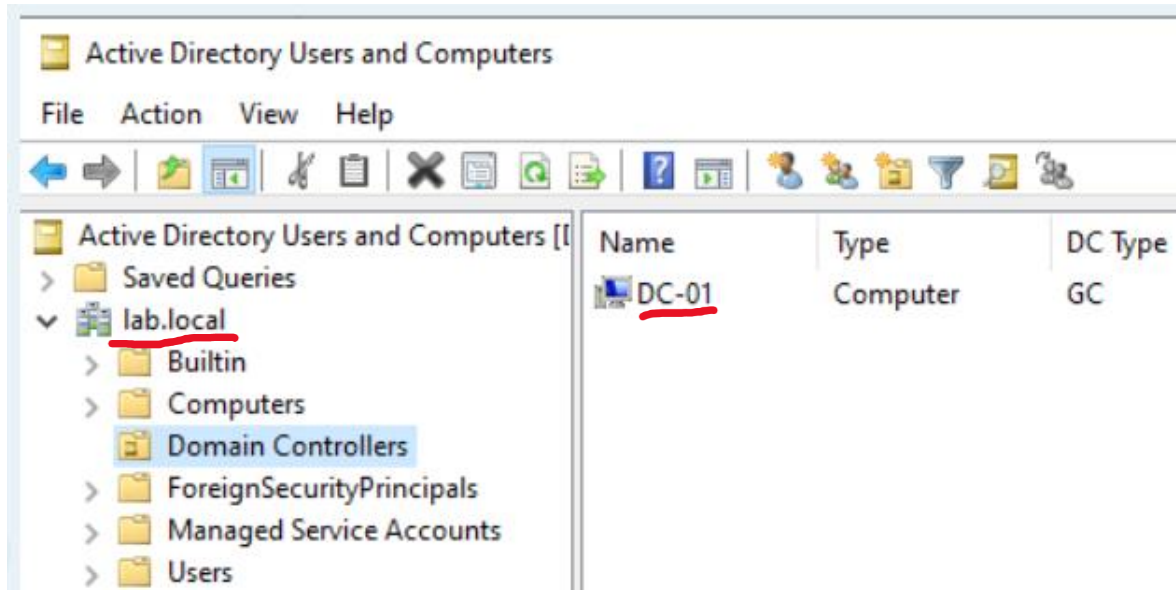


Figure 3: Active Directory Domain / DC View

Phase 3: RSAT & Administrative Tooling

- **RSAT Installation:** Enabled Remote Server Administration Tools via administrative PowerShell by using the command – *Install-WindowsFeature –Name RSAT-ADDS*
- **Directory Management:** Verified functionality of MMC snap-in *dsa.msc* (Active Directory Users and Computers).

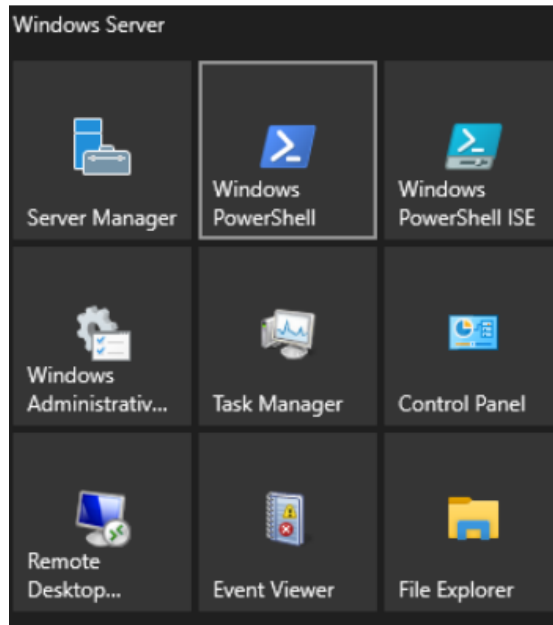


Figure 4: Windows PowerShell

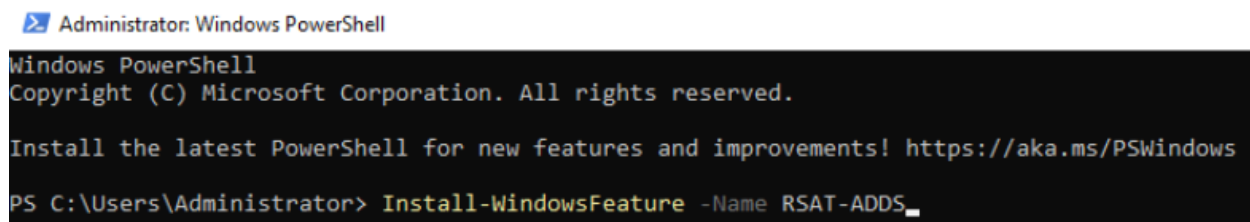


Figure 5: RSAT

Phase 4: Users Directory Provisioning

- Built initial Organizational Unit/Directory structure.
- Provisioned standard test accounts (*lab.local/testuser*) with explicit security policies (custom password complexity, non-expiring password for testing).

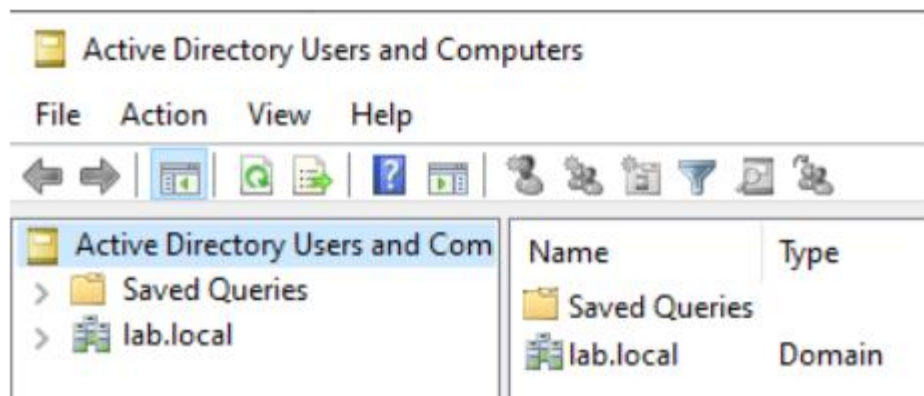


Figure 6: Directory Structure (Domain)

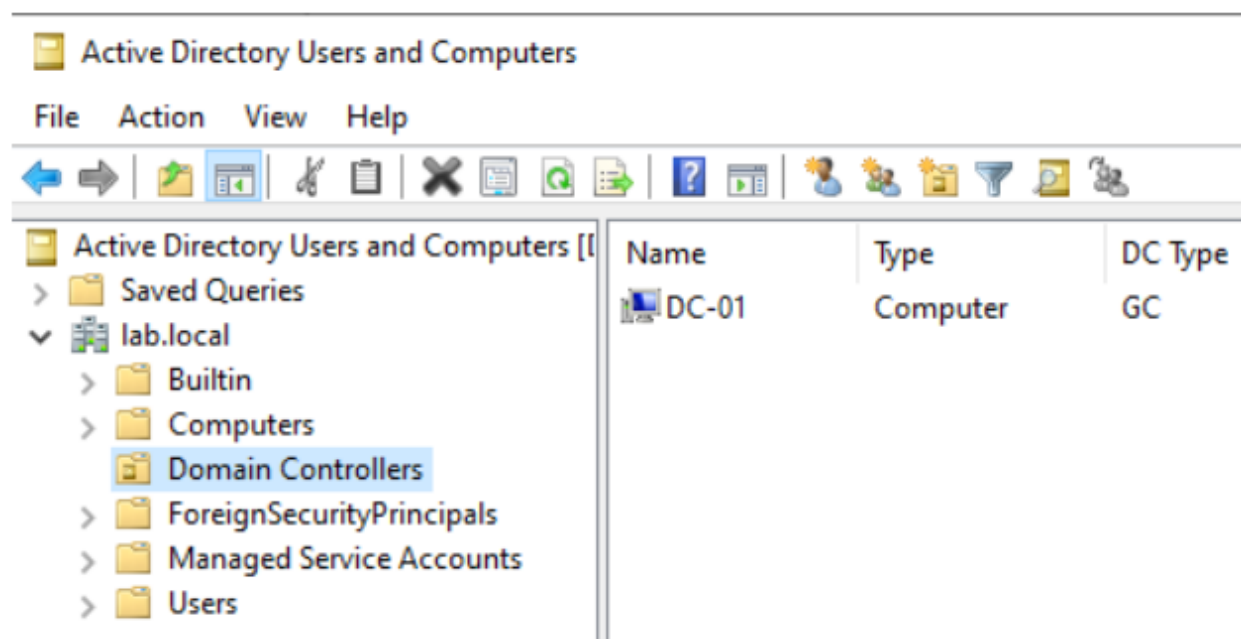


Figure 7: Active Directory Computer

Active Directory Users and Computers

New Object - User

Create in: lab.local/

First name: Initials:

Last name:

Full name:

User logon name: @lab.local

User logon name (pre-Windows 2000):

< Back Next > Cancel

Figure 8: Adding test user

Active Directory Users and Computers

File Action View Help

Name	Type
Builtin	builtinDomain
Computers	Container
Domain Con...	Organizational...
ForeignSecu...	Container
Managed Se...	Container
Users	Container
Test1 T. user1	User

Figure 9: Active Directory User confirmation

System Verification & Results

Test/Component	Status	Verification Method
Prerequisite Check	PASSED	Cleared all networking & credential errors in promotion wizard.
Domain Controller Roles	Active	Successfully authenticated as <i>LAB/Administrator</i> .
DNS Resolution	Functional	Local host successfully resolving domain requests to <i>127.0.0.1</i> .
Directory Object Creation	Successful	Provisioned and validated test domain users within <i>dsa.msc</i> .

Conclusion: This lab successfully demonstrated the end-to-end deployment of a **Windows Server Domain Controller** and the setup of an **Active Directory Domain Services (AD DS)** infrastructure (*lab.local*).

By building this environment, core system administration workflows were validated: setting up foundational network configurations, installing directory roles, troubleshooting system prerequisite failures, managing domains administrative tool via PowerShell, and managing directory objects (Users and OUs). The environment is now fully established and ready for client domain joins, Groups Policy Objects (GPOs), and network services like DHCP/DNS expansions.

Lessons Learned & Technical Takeaway

1. Network Interface Bindings & AD Prerequisites

- **Insight:** Active Directory promotion requires a stable, active network interface. Physical disconnects prevent prerequisite checks from completing successfully.
- **Key Takeaway:** Utilised a software-based **Microsoft KM-TEST Loopback Adapter** to maintain a persistent network binding (*192.168.1.10*) with self-referential DNS (*127.0.0.1*), ensuring directory service stability in isolated lab environments.

2. Administrative Tooling & RSAT Provisioning

- **Insight:** Role installation in Windows Server does not always auto-populate graphical management tools (such as *dsa.msc*) by default.
- **Key Takeaway:** Mastered CLI-based feature installation using **PowerShell** (*/install-
WindowsFeature -Name RSAT-ADDS*) to quickly restore missing management consoles without re-running long graphical setups.

3. Authentication & Account Security Scope

- **Insight:** Promoting a server converts local security accounts into domain-wide identity structures.
- **Key Takeaway:** Learned the distinct differences between local administration (*LAB\Administrator*), along with setting up Directory Services Restore Mode (DSRM) credentials for emergency recovery operations.

4. Enterprise Identity Management

- **Insight:** Directory management relies heavily on clear hierarchy and governance policies.
- **Key Takeaways:** Practised administrative best practices by configuring initial test accounts with structured naming conventions and custom credential management rules suitable for lab testing vs. Production environments.